

How feedback shaped the MVP

DBbun Invisible Illness Burden Simulator. Uri Kartoun, DBbun LLC.

Since the August 13 kickoff I have shown the work at the sprint demos and office hours and changed the product after each round. I held 8 interviews (6 patient advocates from the list of patient advocates the sprint organizers provided for user research, 1 Long COVID patient, 1 government deputy director) and asked 17 people with Lyme on the LymeX Healthathon board a question each; 15 replied. People are described by role, not name. Feedback received does not imply that anyone endorses the product. People who gave feedback are not authors, co-developers or sponsors of the product.

When (2026)	Who	What I heard	What I changed	Where to see it
Aug 13	Federal lead, cost-of-illness track	Which federal resources matter most, and why?	Added a ranked list of the federal sources, each with the reason it ranks where it does (what it lets the tool check or add), not just how often it appears.	Evidence & Sources tab
Aug 13, Sep 10	Federal leads	Show a variety of what-if scenarios and how much results vary, not one answer.	Named scenarios with a stated goal, sensitivity charts, and (Oct 5) a forecast band showing the middle 80% of 500 random draws, labeled as illustrative.	Explore, Sensitivity and Forecast tabs
Aug 21	Federal lead	Comparing Lyme disease and Long COVID could teach each field something ("cross-pollination").	Built the cross-pollination analysis. Since Oct 5 it is a side-by-side comparison: findings with cited simulators, which burdens each paper measures, key figures from each. Thirteen saved comparisons.	Library > Cross-pollination analysis
Aug 31, Sep 3, Sep 9	Long COVID patient; patient advocates	Official counts based on claims and diagnosis codes miss many real cases, and cash-pay costs never appear in claims.	Added a guardrail that flags figures sourced from claims or ICD-10 codes as likely undercounts.	Overview, limitations
Sep 4, Sep 10	Patient-advocate network leader; federal lead	Be clear what is measured, assumed or modeled; show which assumptions drive which numbers; let the reader change them.	Every input is labeled observed, assumed or projected; a "what you changed" panel lists each changed input and the results it feeds; a panel ranks the assumptions that move each result most.	Explore and Overview tabs
Sep 10	Federal leads	Be intentional about sex and gender differences.	The tool discloses sex- or gender-specific findings instead of silently pooling them, in every analysis.	Overview, limitations
Sep 8, Sep 10	Patient advocate; federal lead	Patients fear sharing medical data because of coverage consequences. Review the CMS interoperability framework; how will you handle data governance?	Plain statements of what leaves the machine (a document's text, never the file, reaches the model); every model call recorded in each download; a FAQ answer on federal data standards.	FAQ, NOTICE file in each download
Sep 11	State health-policy official	Needs structured data to feed their own budget forecasting.	Added an Excel export with parameters, formula text, saved values and sources (no recalculation).	Export tab
Sep 20-23	People with Lyme on the LymeX Healthathon board (15 of the 17 I asked replied)	Care split across five or more clinics; a diagnosis delay that turns into years of downstream cost; co-infection costs; years of remote schooling for a child.	Corroborated the uncoordinated-care burden category (added Sept 22 from a published paper); wrote these mechanisms and the schooling example into the generation instructions.	Burden map, burden categories
Sep 24	Federal data stewards, cross-track office hours	The Medical Expenditure Panel Survey withholds three-digit condition codes with fewer than 30 cases and advises caution on Lyme costs.	Evidence cards now say "withheld", not "zero"; the limit is stated in the tool; no Lyme-cost simulator was built from that survey.	Evidence & Sources tab
Sep	AI model reviews of generated bundles	About half of ten review points held up on checking.	Fixed the points that held up in the generation instructions and checks, each with a regression test.	Test suite (227 tests)
Oct 1	Federal lead, Beta Demo	Label time periods and real versus nominal dollars, break cost out by who pays, and signal where data are incomplete.	Added price-year and data-period labels, a "who bears the cost" table, and a burden map showing which of 16 burdens a paper measured and which it left out.	Explore and Overview tabs
Oct 1	Patient advocates, Beta Demo	Make the choice of audience easier to see; offer an audio option.	Moved the audience choice above the tabs. Built a spoken overview (synthetic voice) with transcript and MP3 download for every simulator and comparison. I had planned video too and dropped it.	Library rows; comparison pages
Oct 8	Federal statistics experts	Long COVID "ever" and "currently" are different measures and surveys differ; the national health survey has never asked about Lyme.	The submission states there is no Lyme question in the survey codebooks. Planned after Oct 15: refresh the Long COVID evidence cards and label "ever" and "currently".	Submission text; planned
Oct 8	Federal regulator expert; insurance-program data staff	Software that organizes information should stay associative, be labeled as an aid, and prompt human review. No public source ties claims to Lyme or Long COVID codes.	Removed "decision-support" wording from the product text; stated it gives no medical advice and recommends no treatment; Blue Button 2.0 is shown only against the federal sandbox.	FAQ, limits, standards answer
Sep 8, Oct 8-10	Patient advocate; community members testing the public demo	"Simulator" meant nothing; the page is too complicated and text-heavy; wanted a one-line summary per study, a condition filter and a picture of how it works.	Added a "what this is and is not" banner, one-line study summaries, a Lyme / Long COVID filter, a collapsed coverage map, FAQ step graphics and pictures of the interactive tool.	Library, FAQ, banner
Sep 3 to Oct 8	Advocates; federal leads (heard, not built yet)	Household income hides individual access; mold and air-quality costs; caregiver substitution; a "who loses coverage" scenario; a two-way way to flag assumptions that do not match reality.	Not built. The household-income caveat is deliberately left open. Unmeasured burdens are shown as gaps on the burden map, not estimated.	Burden map

AI tools used. In the product: Claude Sonnet 5 (model ID claude-sonnet-5, Anthropic) reads uploaded papers and writes each simulator's specification and wording; code validates it and does all arithmetic. To build and test the software: Claude Code with Claude Sonnet 5.5 (claude-sonnet-5.5). AI reviews of generated simulators and code: GPT and GPT-5 Codex (OpenAI; exact builds not exposed), Grok 4.5 (xAI), Gemini 3.5 (Google) and Claude Sonnet 5.5 (Anthropic). Audio overviews: scripts written with Claude Code; voice by Kokoro-82M, an open-source text-to-speech model run locally.